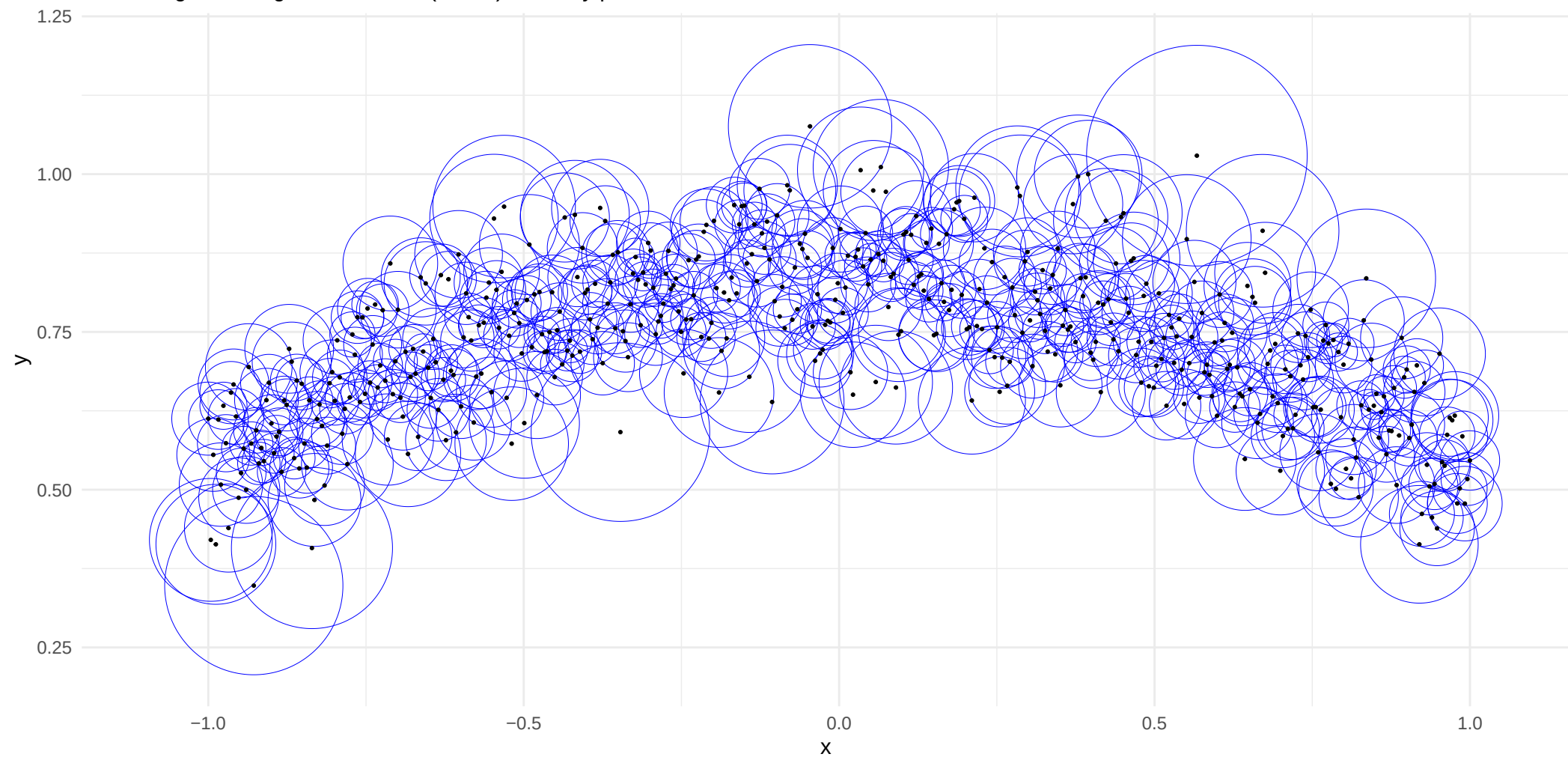
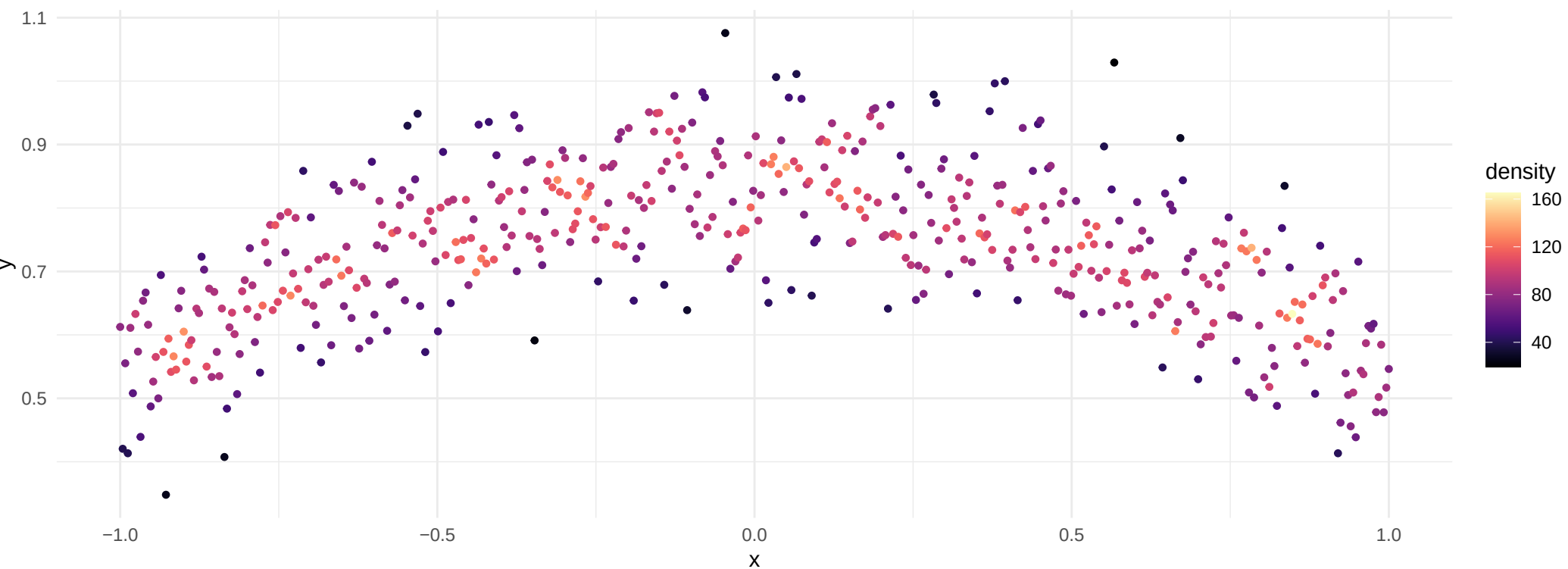


All Adaptive Spheres

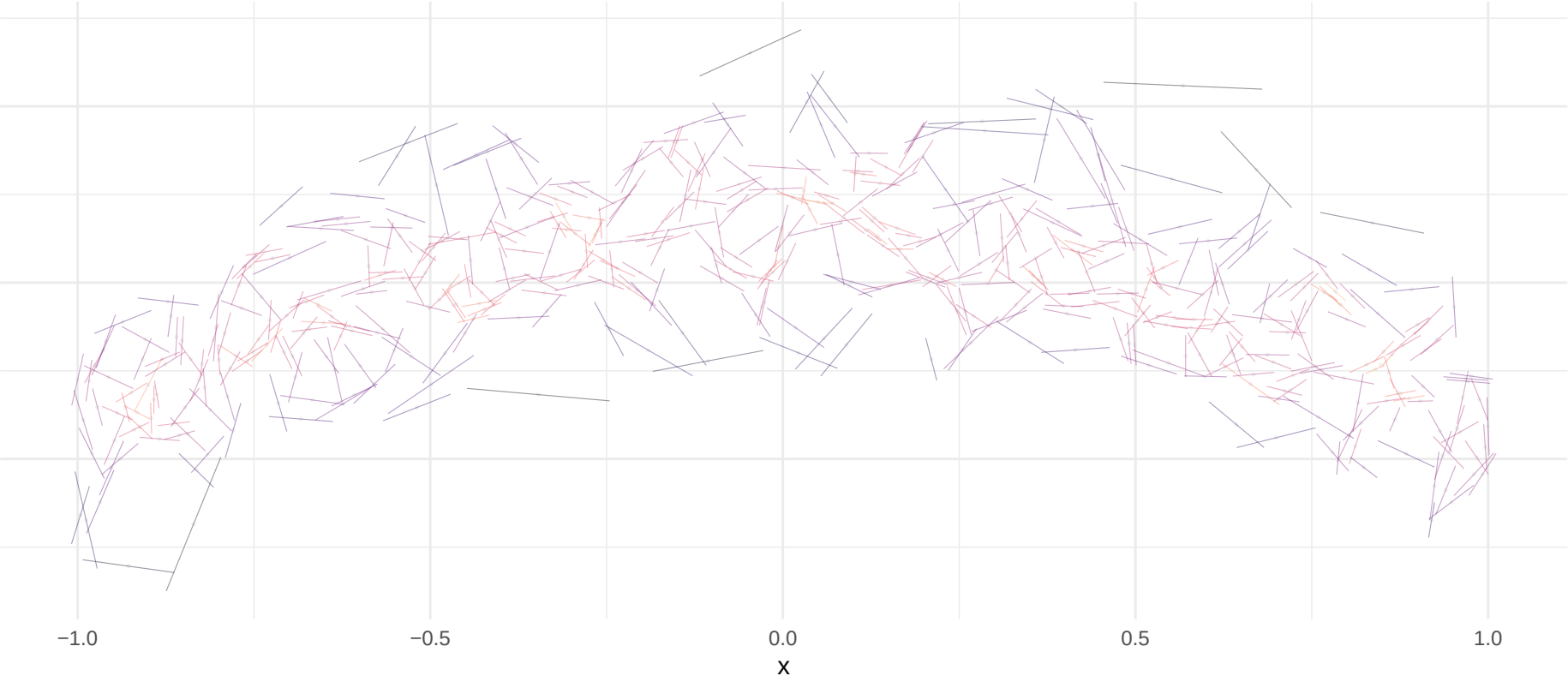
Visualizing local neighborhood size (radius) for every point.



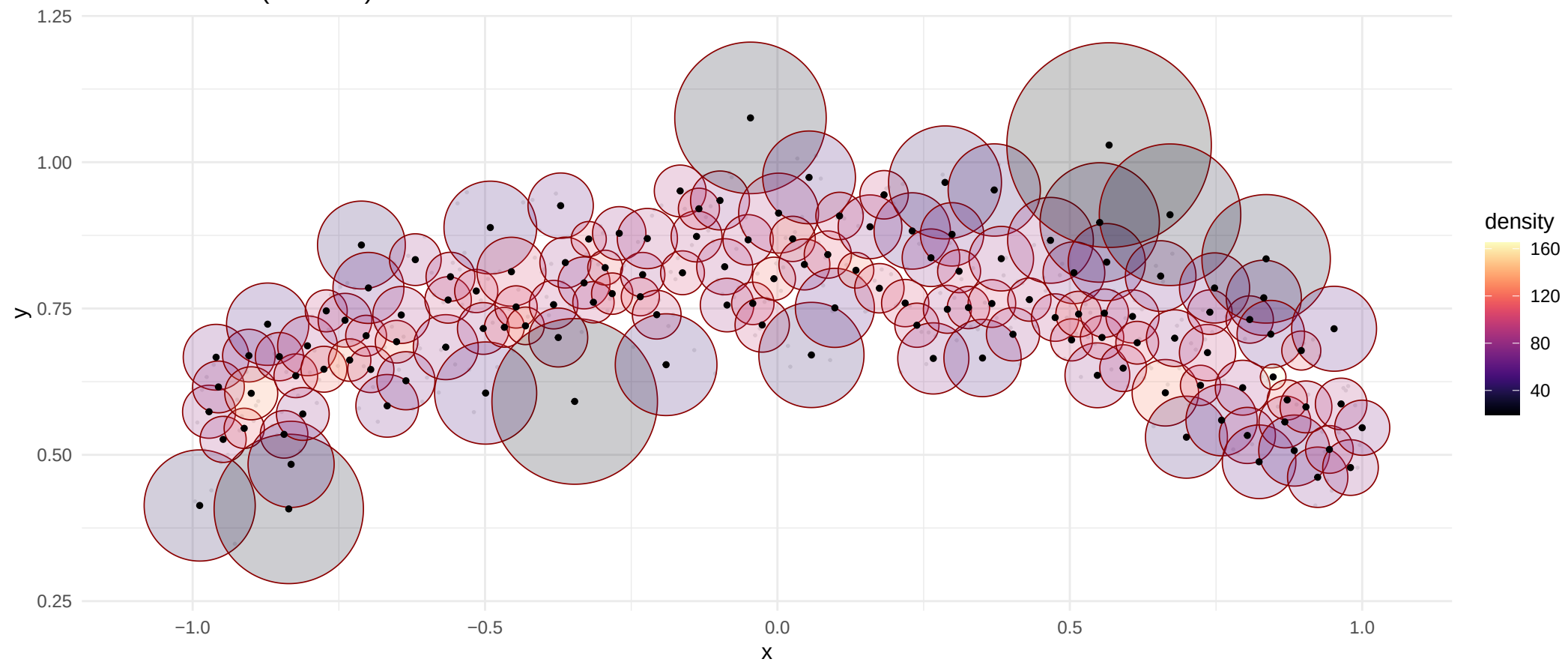
Point Density Distribution



Global Tangent Field (PC1)

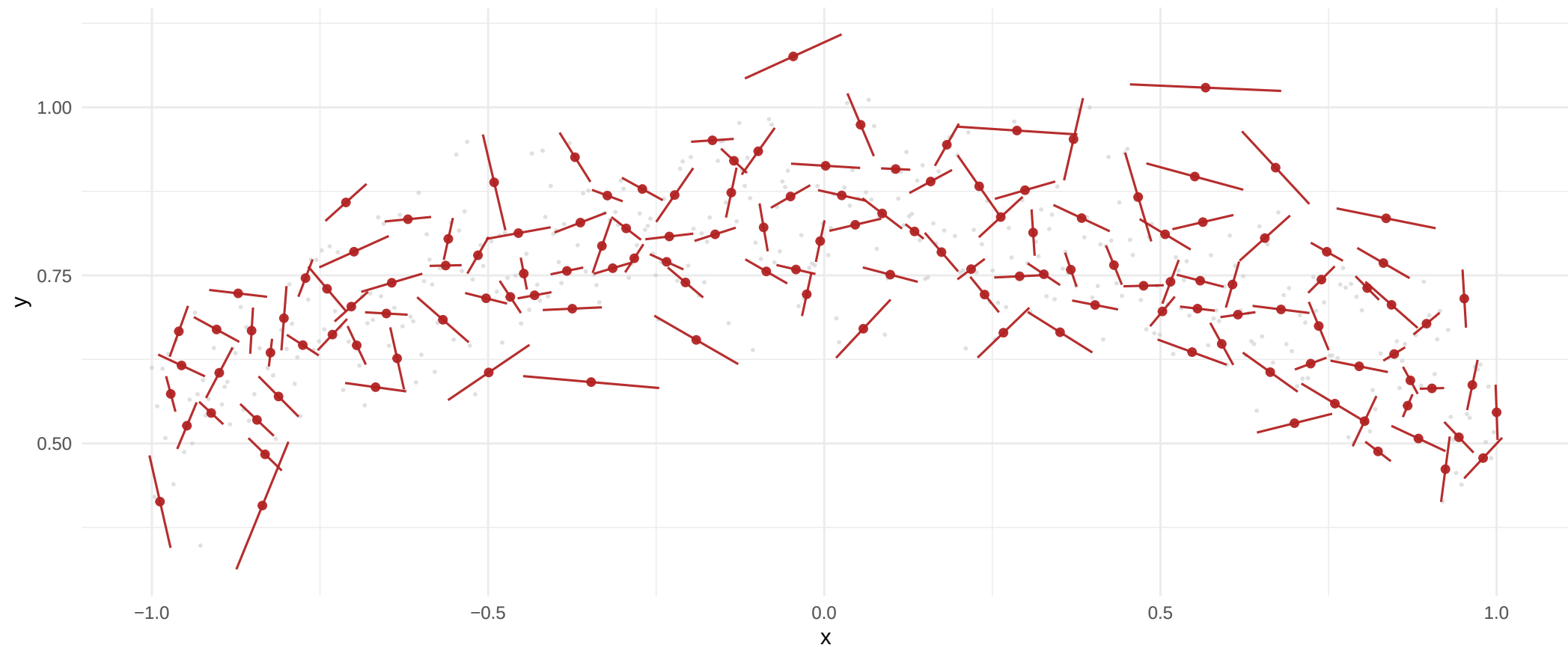


Atlas Selection (Anchors)



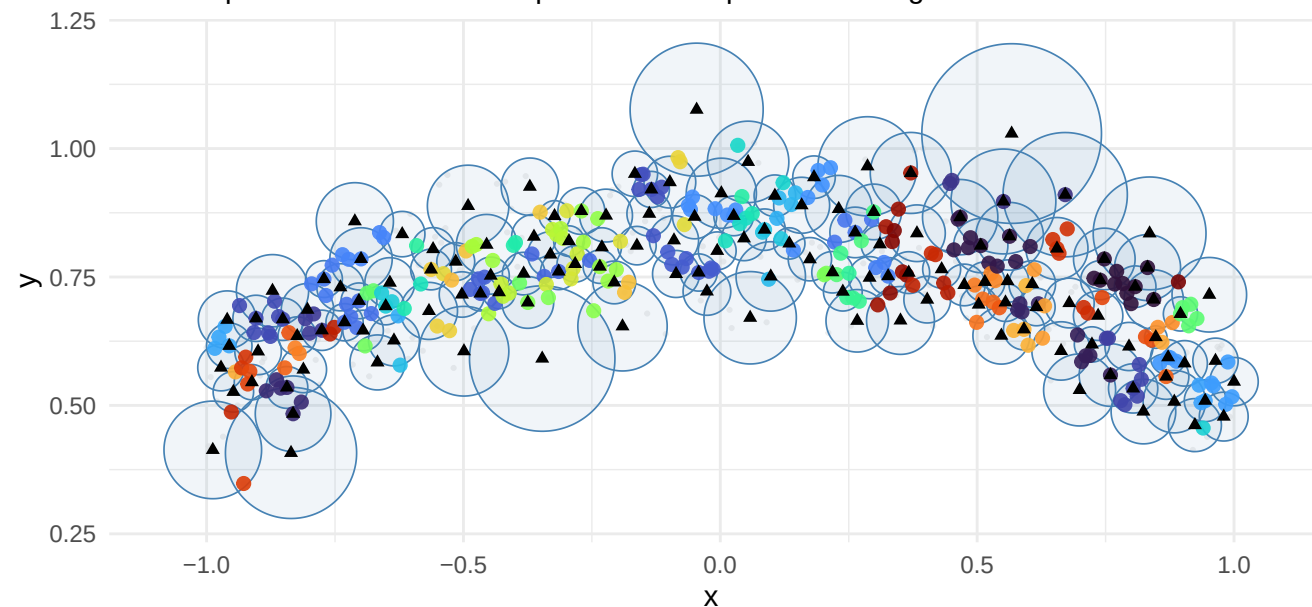
Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



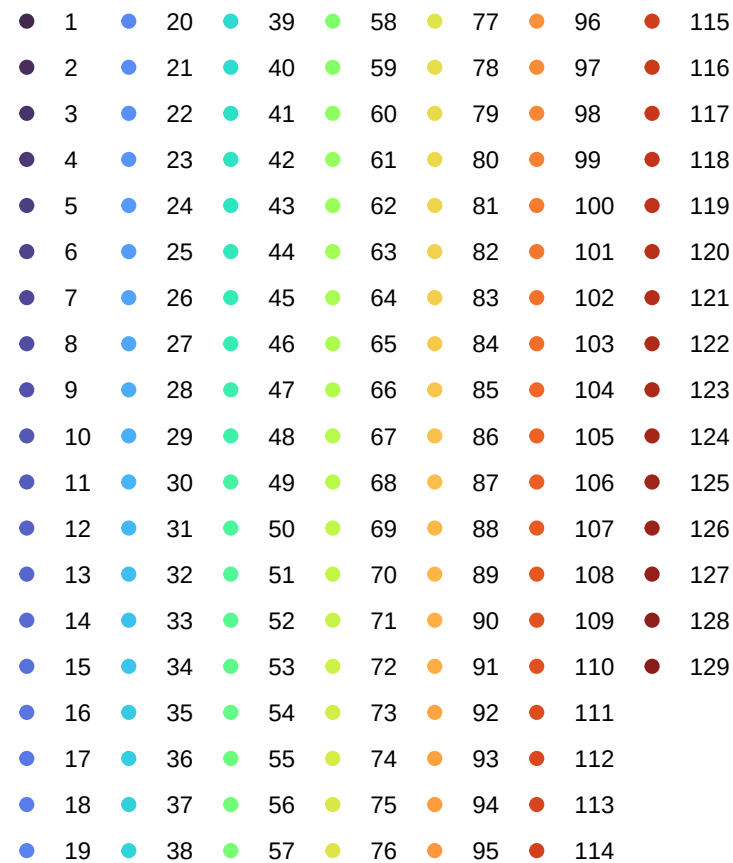
Intersection Regions (2+ anchors)

Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.



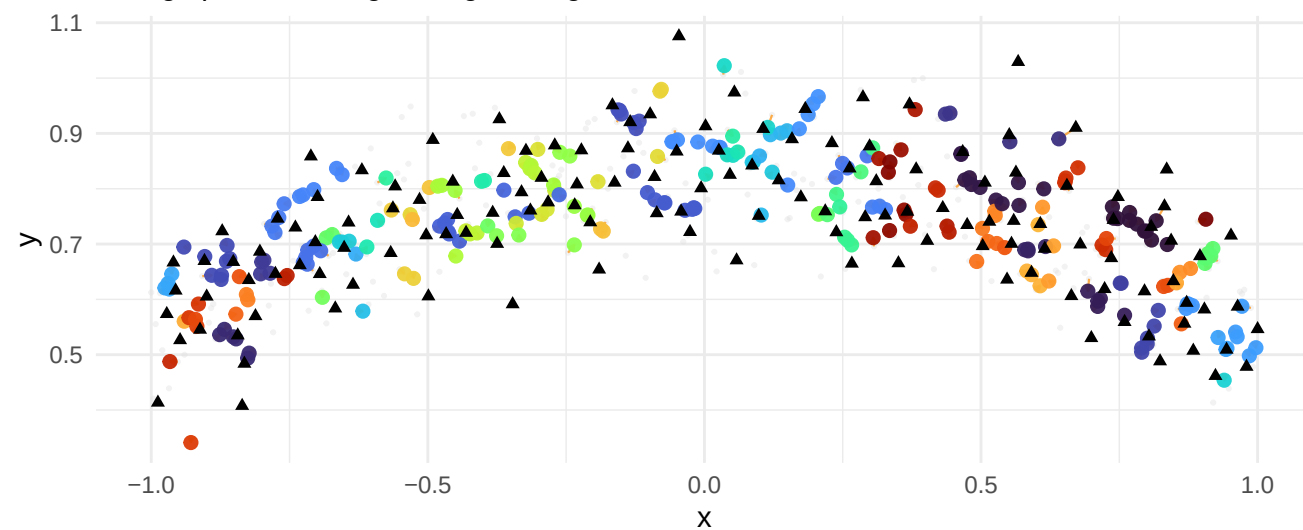
k_min = 5, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01

Region



Stitched Backbone (Iteration 1)

Orange path: stitching; Triangles: original anchors

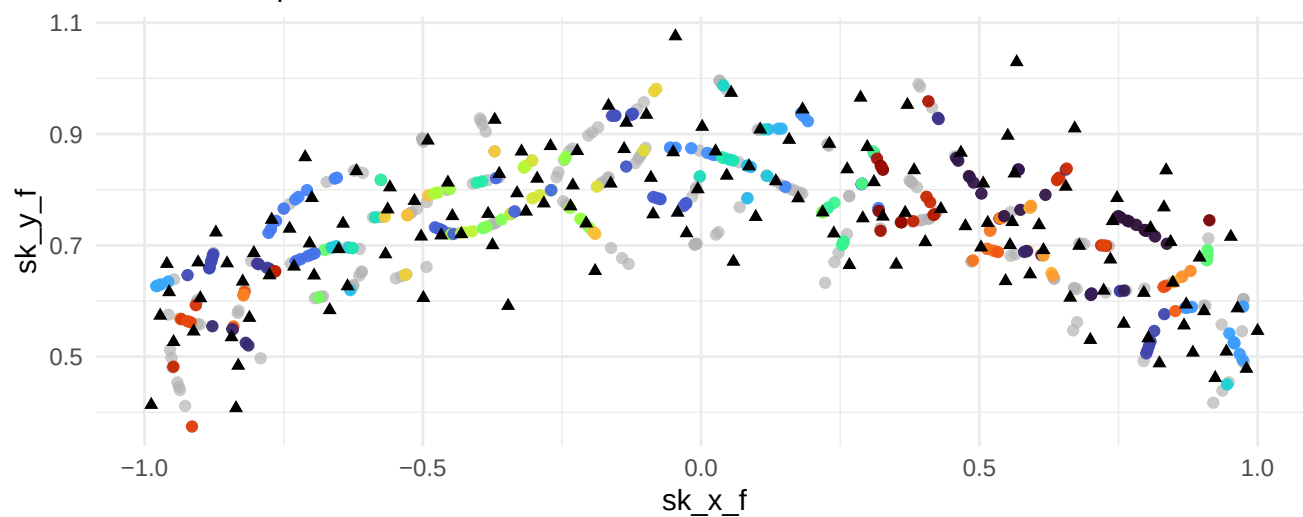


Region

1	20	39	58	77	96	115
2	21	40	59	78	97	116
3	22	41	60	79	98	117
4	23	42	61	80	99	118
5	24	43	62	81	100	119
6	25	44	63	82	101	120
7	26	45	64	83	102	121
8	27	46	65	84	103	122
9	28	47	66	85	104	123
10	29	48	67	86	105	124
11	30	49	68	87	106	125
12	31	50	69	88	107	126
13	32	51	70	89	108	127
14	33	52	71	90	109	128
15	34	53	72	91	110	129
16	35	54	73	92	111	
17	36	55	74	93	112	
18	37	56	75	94	113	
19	38	57	76	95	114	

Final Stitched Backbone (Converged)

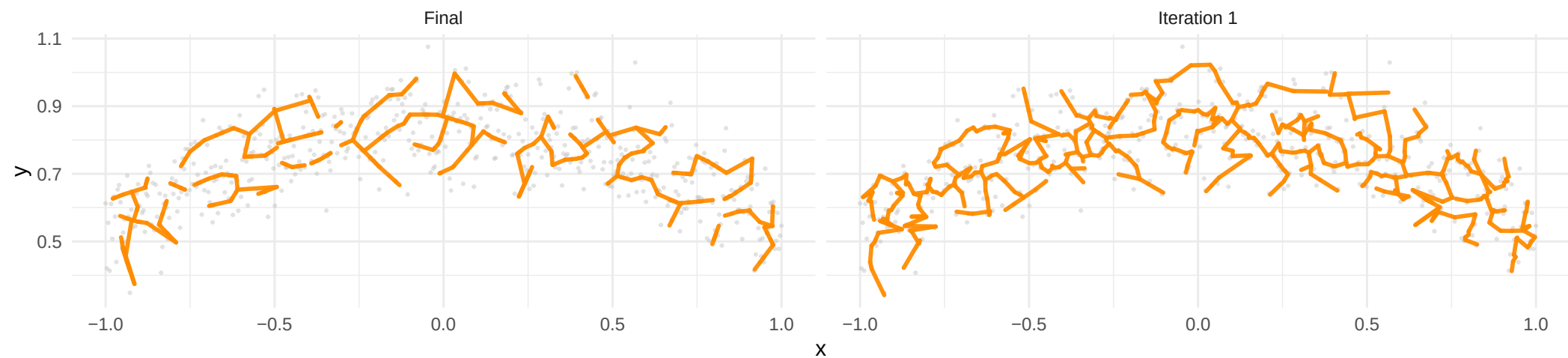
Final stitched points



Region

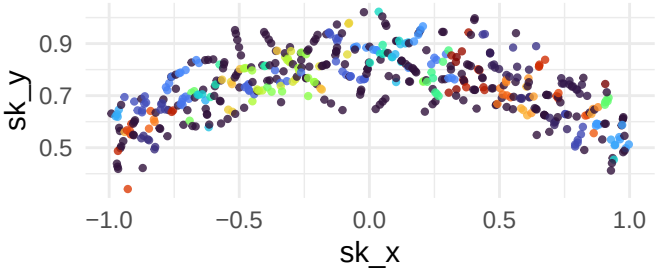
1	20	39	58	77	96	115
2	21	40	59	78	97	116
3	22	41	60	79	98	117
4	23	42	61	80	99	118
5	24	43	62	81	100	119
6	25	44	63	82	101	120
7	26	45	64	83	102	121
8	27	46	65	84	103	122
9	28	47	66	85	104	123
10	29	48	67	86	105	124
11	30	49	68	87	106	125
12	31	50	69	88	107	126
13	32	51	70	89	108	127
14	33	52	71	90	109	128
15	34	53	72	91	110	129
16	35	54	73	92	111	
17	36	55	74	93	112	
18	37	56	75	94	113	
19	38	57	76	95	114	

Backbone Edges

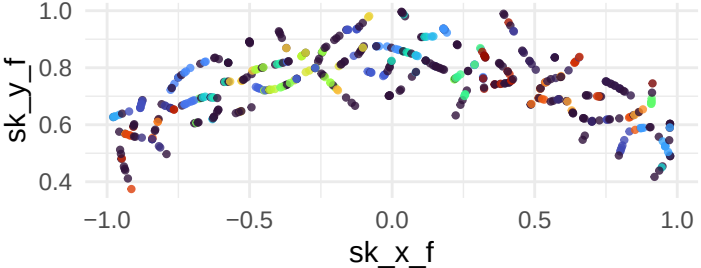


Convergence: iteration 1 vs final

Iteration 1

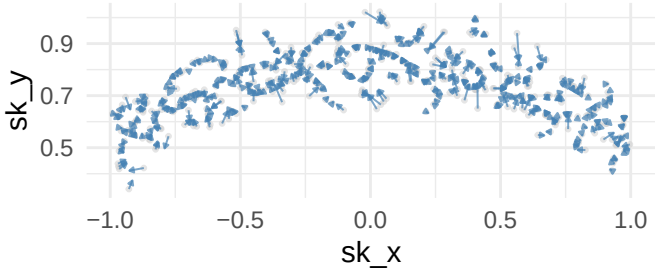


Converged



Displacement iter1 -> final

Points moved > 1e-6: 500 / 500



Parameters: k_min = 5, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01